

Muhammad Daffa Raihan

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Professional Summary

Undergraduate Informatics student with strong full-stack development experience across backend, frontend, and mobile platforms. Successfully built and shipped multiple end-to-end systems including the IMUII automation platform and HalalMate, an award-winning AI-powered mobile application. Proficient in Golang, Node.js, React, and Flutter to deliver scalable and user-centric digital solutions.

Skills & Tools

Backend: Golang, Node.js (NestJS, Express)
Frontend: React, Vite, Tailwind CSS, HTML/CSS
Mobile Development: Flutter, Jetpack Compose
Databases & Caching: PostgreSQL, MySQL, Redis
AI & Data Integration: OpenAI API, OCR, Web Scraping, XAI Integration
DevOps & Tools: VPS, Docker, Railway, Git, Postman

Projects

IMUII	
Founder & Technical Architect	Des 2025 - Present
- Live Platform: https://imuii.id Management Portal: https://portal.imuii.id - Architected and built IMUII, an end-to-end deployment automation platform supporting 45 projects with an 82% successful deployment rate across multiple environments. - Designed Go-based backend services and CLI tooling enabling one-command project initialization, automated CI/CD pipeline configuration, and environment provisioning. - Implemented GitHub repository automation, zero-config domain provisioning, and real-time deployment monitoring for 47 active subdomains via a centralized management portal. - Ensured infrastructure efficiency and reliability through optimized resource management, maintaining low CPU usage (~1%) under active deployment workloads.	
HalalMate	
Full Stack Mobile Developer	Feb 2025 - Jan 2026
- Developed HalalMate, an AI-powered mobile application to assist Muslim travelers in verifying halal food options by reducing language barriers, technical ingredient codes, and interpretation complexity. - Built Golang-based backend services to aggregate and process restaurant and product data, including automated data extraction from Google Maps using ChromeDP. - Implemented an AI-driven halal detection pipeline using OCR and OpenAI, achieving 93% classification accuracy with 100% recall (zero false-negative cases), ensuring reliable halal verification. - Developed the Flutter mobile application with location-based search, map integration, and filtering features to enhance user experience in discovering nearby halal options.	
KeepUp	
Backend Developer	Aug 2024 - Jun 2025
- Developed backend services for a faculty-led mental health screening research project implemented at Universitas Islam Indonesia, supporting 200+ student users. - Designed and implemented RESTful APIs using NestJS, PostgreSQL, and Redis to manage questionnaires, process screening results, and handle concurrent user access. - Implemented role-based authentication and authorization to secure sensitive mental health data for administrators and student users. - Built automated reporting features generating PDF and Excel outputs, integrating OpenAI to produce comprehensive and interpretable mental health assessment reports.	
TikaTiko	
Back-End Developer	Sep 2024 - Nov 2024
- Developed a backend system to manage laboratory assistant attendance and automate weekly recapitulation for informatics laboratories. - Designed and implemented database schemas to store attendance records, schedules, and reporting data in a structured and reliable manner. - Integrated a Telegram bot interface to streamline attendance submission and deliver summarized weekly reports to laboratory managers. - Implemented automated report generation to reduce manual administrative workload and improve data accuracy for laboratory operations.	
Nasho	
Back-End Developer	Aug 2024 - Oct 2024
- Developed backend services for an Arabic language learning platform featuring structured learning paths and quiz-based progression. - Implemented progress tracking and a content lock system that unlocks learning materials only after users complete required quizzes. - Designed and built RESTful APIs using Express.js and PostgreSQL to support user authentication, learning content delivery, and progress analytics. - Implemented role-based access for administrators to manage learning materials, quizzes, and curriculum structure.	

Work Experience

Emica	
Full Stack Web Developer	Jun 2025 - Present
- Developed and maintained full-stack web applications for client-facing products, including a restaurant reservation and analytics system used in production. - Designed and implemented backend APIs and database schemas using Node.js and PostgreSQL to handle reservations, customer data, and analytics processing. - Built web interfaces using React to visualize business metrics, including visitor traffic, reservation trends, and customer behavior. - Implemented WhatsApp booking and analytics features that achieved an 85% click-to-contact conversion rate and supported 3,900+ reservations from real users.	

Fakultas Teknologi Industri	
Assistant Laboratory on Informatics	Sep 2023 - Jan 2026
- Assisted in teaching and mentoring undergraduate students across 7+ classes in core computer science courses, including Programming Logic, Web Development, Algorithms & Data Structures, and Computer Networks & Systems. - Supported 150+ students by providing hands-on guidance in problem-solving, coding fundamentals, networking concepts, and application development. - Assisted in lab sessions, assignments, and technical evaluations to ensure students met learning objectives and programming standards. - Strengthened communication and leadership skills by translating complex technical and networking concepts into clear and practical explanations.	

SaveBite	
Full Stack Mobile Developer	Aug 2024 - Jan 2025
- Developed backend services for a food-related mobile application, enabling data exchange between mobile clients and server-side systems. - Built RESTful APIs using Node.js and Express to support core application features and data persistence. - Designed and implemented relational database schemas to support Android application workflows. - Developed Android mobile features using Jetpack Compose to deliver a modern and responsive user interface.	

TEDXUI	
Back-End Developer	May 2024 - Jun 2024
- Developed backend APIs for a web-based ticketing system to manage product listings and order processing. - Implemented RESTful services using Express.js, PostgreSQL, and Redis to handle transactional workflows. - Built administrative APIs to manage products and ticket inventory. - Deployed the backend application on a VPS environment to support production usage.	

Education

Islamic University of Indonesia	
Informatics Engineering	Aug 2022 - Aug 2026
3.96 GPA - Member of Programming Study Club, 2023 to 2024 - Member of Google Development Students Club, 2023 to 2024	

Man Insan Cendekia Tanah Laut	
High School Diploma	Aug 2019 - Aug 2022
88/100 GPA - Member of Mathematics Club, 2020 to 2021	

ACHIEVEMENTS

Best Paper Award Automata 2025	
Informatika Universitas Islam Indonesia	Des 2025
- Paper: An XAI-Based Halal Detection System for Packaged Food Products Integrating Vision Models and Web Retrieval - Developed an explainable AI (XAI) system for halal detection on packaged food using vision models and web-based information retrieval.	

Silver & Bronze Medal on INDES 2025	
UiTM Perak	Jun 2025 - Jul 2025
- Silver Medal – “HalalMate” Mobile App (AI-Based Halal Food Recommender) - Built with Flutter and integrated an AI model to recommend halal food options based on user location and preferences. - Implemented search filtering, user preferences, and location-based results using Maps API. - Developed full AI pipeline: scraped food data, built halal-haram classifier, integrated OCR for label reading, and AI-powered chatbot for user interaction. - Bronze Medal – “Goqu” Web App (Donation Management System) - Developed a donation transparency platform for mosques using React, Tailwind, and Express JS. - Implemented user authentication, real-time fund tracking, and admin dashboards. - Designed for scalability and mobile responsiveness, targeting small to mid-size masjid operations.	